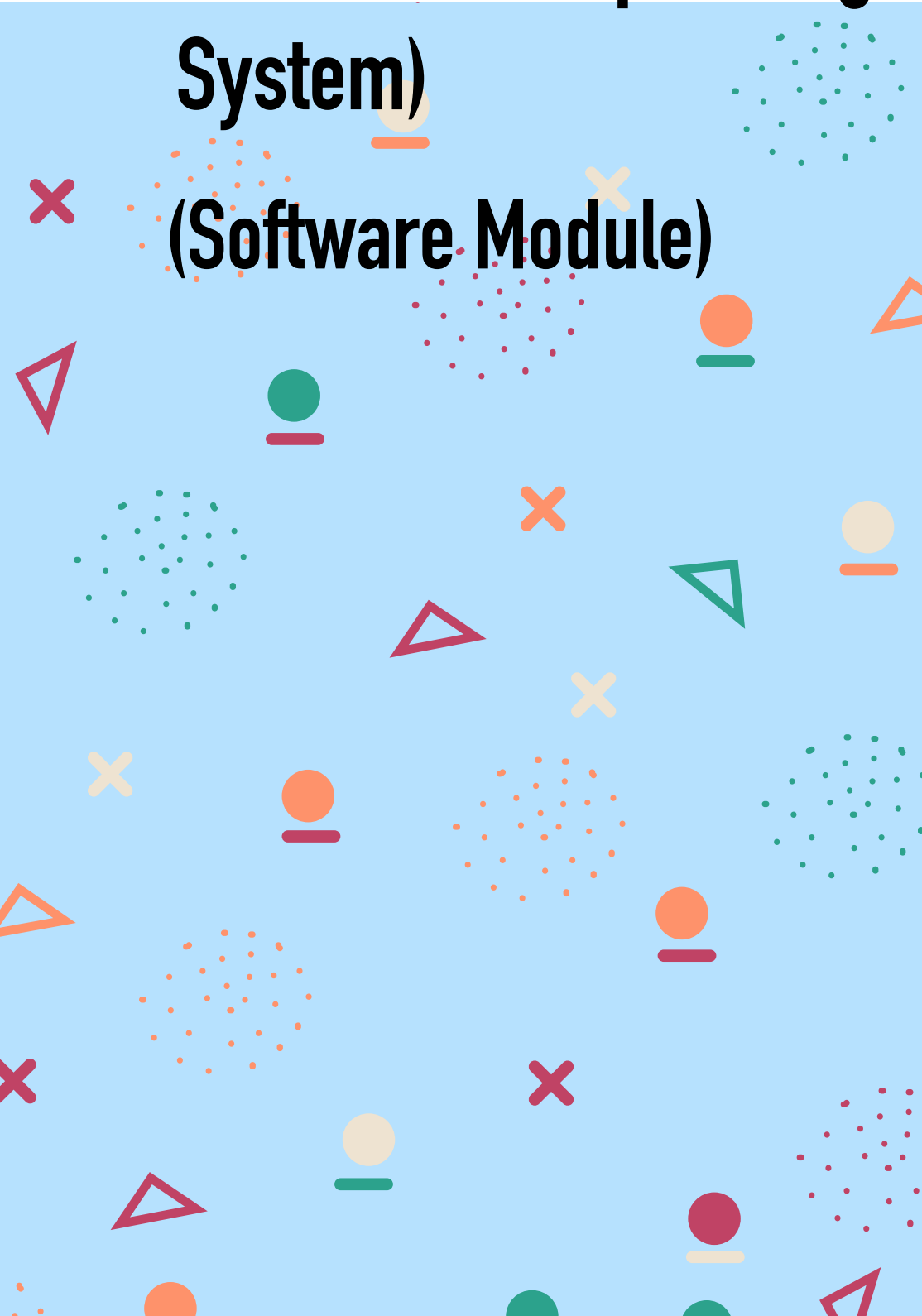


ROS (Robot Operating System)

(Software Module)



ROS2 is a set of libraries & fns to make robots

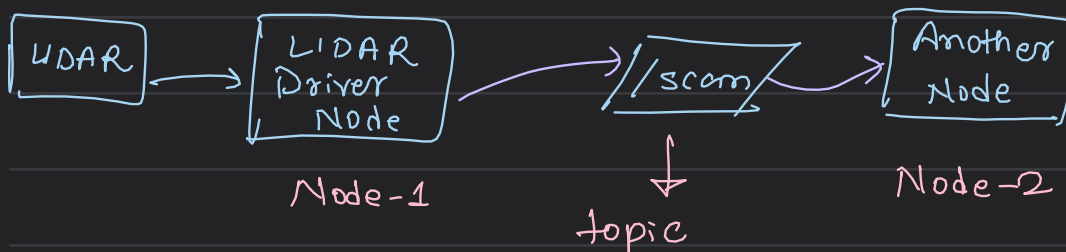
- ① Nodes: bunch of smaller Programs that all run at the same time & communicate with each other. (uses ROS2 Libraries)
(written any Comp. lang.)

Command to run a node:

```
ros2 run package-name node-name
```

How nodes communicate with each other? through topics & services

- ② Topics: it is a named location where a node can publish a message (called publishers) & then other nodes called Subscribers can subscribe to this topic & read the message.
* A node can be a publisher for 1 message & a subscriber to another message from same or other topics.



Useful commands while working with a topic

- ① topic list:

Useful commands while working with topics belonging to all the nodes
`ros2 topic list`

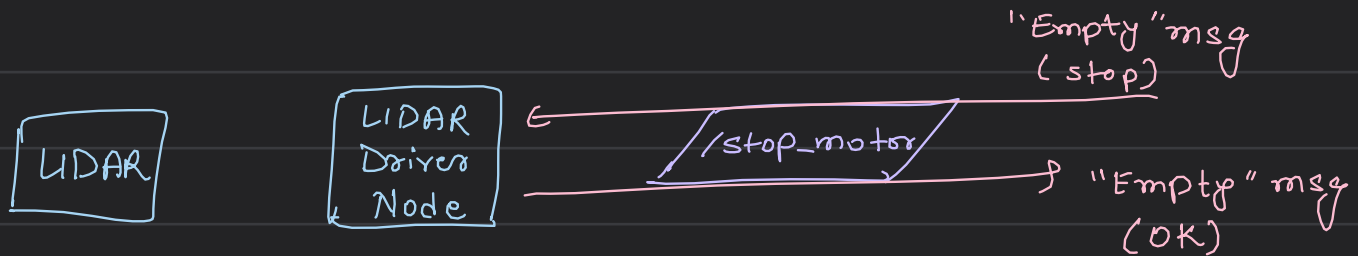
to see the topics of a particular node which it either publish or subscribe

```
ros2 node info / node-name
```

- ② echo: to pretend as a subscriber & see what the topic is publishing (message)
`ros2 topic echo /scan --no-arr`

③ **Services**: similar to topics but used when we require single message request-reply communication

to send a particular piece of info & getting some other info in return



Useful commands for a service

① **List**:

`ros2 service list`

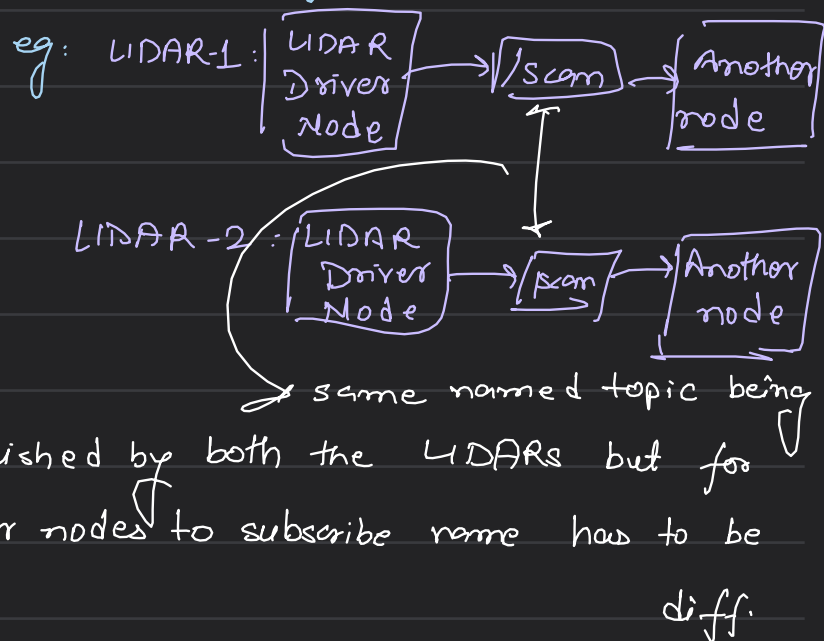
eg: `/start_motor`
`/stop_motor`

`ros2 service call /stop_motor` } to call a particular service
↳ will stop the motor & say "OK"

④ **Parameters & Remapping**: to change the name of the running topics / services

to change the behavior of the node

(eg: we have 2 LIDARs instead of 1 as above → node needs to know which LIDAR is giving which signal)



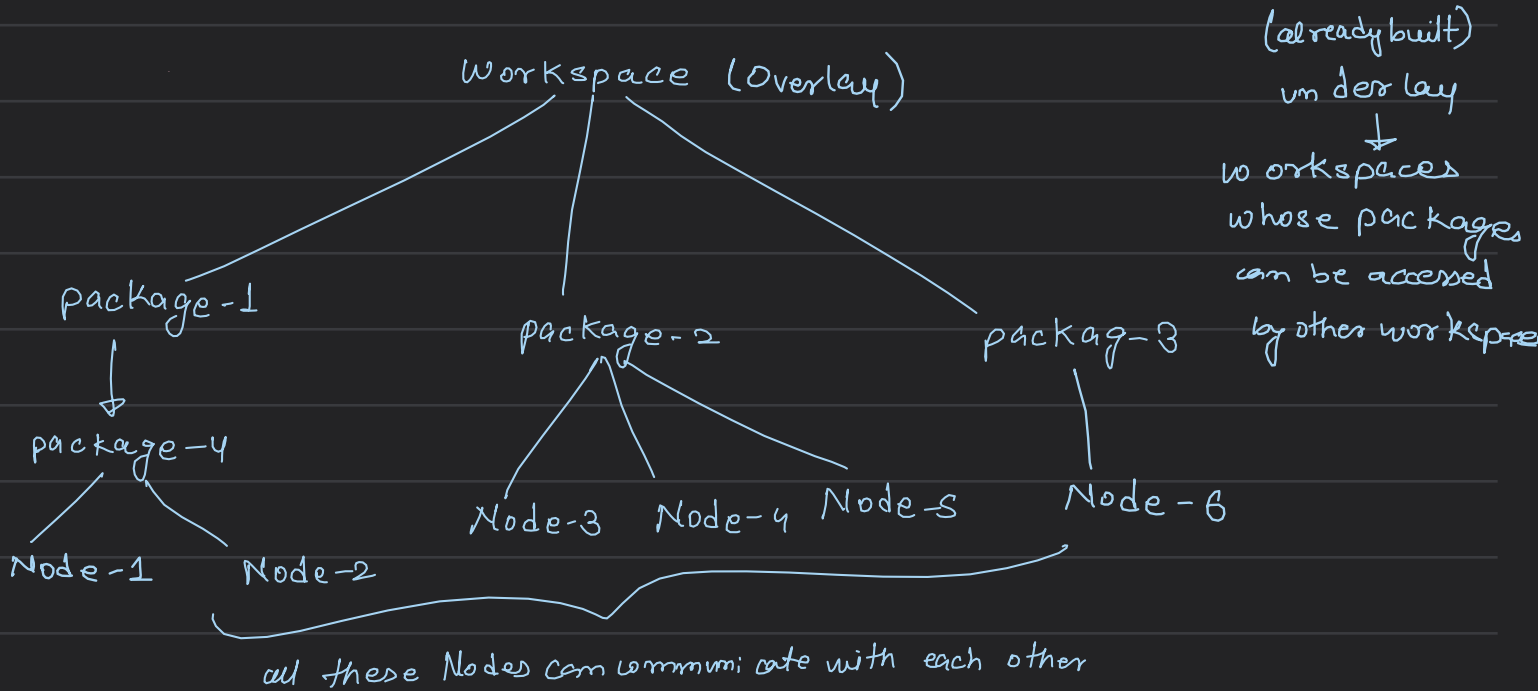
⑤ Launching: to launch multiple nodes together of a package

```
1 from launch import LaunchDescription
2 from launch_ros.actions import Node
3
4 def generate_launch_description():
5     return LaunchDescription([
6         Node(
7             package='rplidar_ros',
8             executable='rplidar_node',
9             parameters=[
10                 'serial_port': '/dev/ttyUSB0',
11             ],
12             remappings=[
13                 ('/scan', '/scan_1'),
14             ]
15         ),
16         Node(
17             package='rplidar_ros',
18             executable='rplidar_node',
19             namespace='/scanner2',
20             parameters=[
21                 'serial_port': '/dev/ttyUSB1',
22             ]
23         )
24     ])
```

— using launch we are running
2 nodes simultaneously here

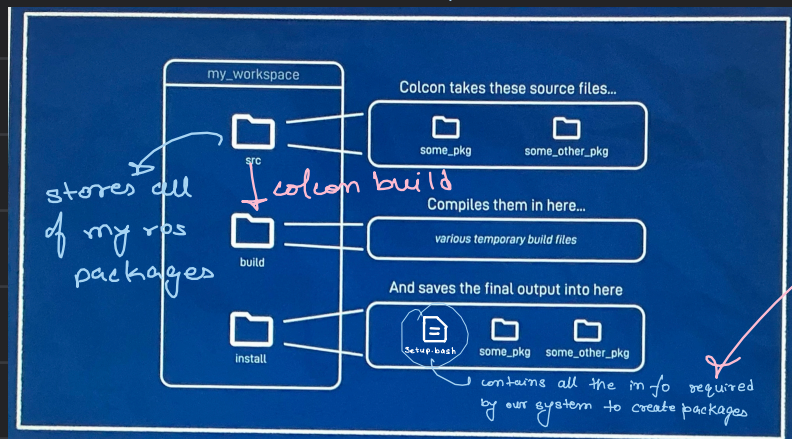
run → launch ⇒ all the nodes of the
package starts running

(just an example)



Using colcon to build workspace:

↳ a build manager: its job is to find all ROS packages in workspace, compile them & prepare them so ROS can run them.



source install / setup-bash

↓
at the start of every
terminal

due to
this

① Creating a Workspace:

command: `mkdir -p ros2_ws/src`

`cd ~/ros2_ws/src` we enter into this workspace

Note: `ls` command to list all the files in a workspace/package/folder...

② Creating a Package: (a package is created inside the `src`)

Command: `ros2 pkg create --build-type ament_cmake package-name`